

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech.

Branch: Civil Engineering

Semester: VII

Subject Code & Name: BTCVC701 & Design of Reinforced & Prestressed Concrete Structures

Max Marks: 60

Date:02-01-2024

Duration: 3 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.
5. Use of IS 456-2000, IS 1343-2012 is permitted.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

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|---|----------|----------|
| A) Define torsion and derive equation for torsional stiffness for a member subjected to torsion. Assume suitable nomenclature. | (L2/CO1) | 6 |
| B) A reinforced concrete rectangular beam $b = 300$ mm, $d = 500$ mm and $D = 550$ mm is subjected to factored shear force $V_u = 75$ kN, $M_u = 170$ kNm, $T_u = 50$ kNm. Assume M 20 concrete, Fe 415 steel. Design the beam for limit state of collapse in torsion. (Skip transverse steel calculation step) | (L3/CO1) | 6 |
| C) A reinforced concrete rectangular beam of $b = 300$ mm, $d = 500$ mm and $D = 550$ mm subjected to factored shear force of 100 kN. Assuming % age of steel as 1% in the section, determine the factored torsional moment in the section if maximum steel for torsion is provided in the section. Assume M 30 grade concrete and Fe 415 Steel. | (L3/CO1) | 6 |

Q.2 Solve Any Two of the following.

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|---|----------|----------|
| A) Explain the classification of column based on slenderness ratio. Also, state the provisions for longitudinal reinforcement and transverse reinforcement. | (L2/CO2) | 6 |
| B) Design the reinforcement in a column of size 500 mm x 700 mm subjected to an axial load of 2100 kN under service load. The column has an unsupported length of 3.0 m and effectively held in position and restrained against rotation at one end but not held in position nor restrained against rotation at the other end. Use M 25 concrete and Fe 415 Steel. | (L3/CO2) | 6 |
| C) Design a short column subjected to a factored load of 1400 kN and factored moment of 90 kNm about minor axis (y-y). The column has a length of 5 m with one end hinge and other end fixed. Use M25 grade concrete and Fe415 steel. | (L3/CO2) | 6 |

Q. 3 Solve Any Two of the following.

- A) Differentiate between pre-tensioning and post-tensioning methods of prestressed concrete. (L2/CO3) 6
- B) Explain with neat sketch 'Magnet Blaton system' of pretensioning. (L2/CO3) 6
- C) What are the advantages of prestress concrete over RCC? (L2/CO3) 6

Q.4 Solve Any Two of the following.

- A) List and explain the types of losses in the prestressed (pre-tensioned and post-tensioned) concrete beam. (L2/CO3) 6
- B) A prestressed concrete beam of rectangular section 300 mm wide and 600 mm deep has a span of 15 m. The effective force is 1520 kN at an eccentricity of 150 mm. The DL of the beam is 5 kN/m and LL on beam is 8 kN/m. Determine the extreme stresses in concrete at i) end span, ii) mid span. (L3/CO3) 6
- C) A post-tensioned beam 150 mm × 400 mm (b × h) spanning over 10 m. Each cable has a parabolic profile with a dip of 50 mm at mid-span and having a cross-sectional area of 250 mm². It has initial stress of 1200 MPa. If the cables are tensioned from one end, estimate the percentage loss in the cable due to friction at the anchored end. Assume $\mu = 0.35$, $k = 0.0015/\text{m}$. (L3/CO3) 6

Q. 5 Solve Any Two of the following.

- A) What is structural auditing? What are its advantages and how it is carried out? Explain. (L2/CO4) 6
- B) Design a post-tensioned hanger to carry an axial tension of PDL = 250 kN (dead load including self-weight) and PLL = 150 kN. The dimension of the hanger is 300×300 mm². Design the section without considering non-prestressed reinforcement. Tension is not allowed under service loads. The grade of concrete is M 35. The age at transfer is 28 days. Assume 15% long term losses in the prestress. The following properties of the prestressing strands are available from tests. Type of prestressing tendon : 7 wire strand, nominal diameter = 12.8 mm, nominal area = 99.3 mm², tensile strength (f_{pk}) = 1860 N/mm², modulus of elasticity = 195 kN/mm². (L3/CO4) 6
- C) Design a simply supported Type 1 prestressed beam with MT = 435kNm (including an estimated MSW = 55 kNm). The height of the beam is restricted to 920 mm. The prestress at transfer $f_{po} = 1035 \text{ N/mm}^2$ and the prestress at service $f_{pe} = 860 \text{ N/mm}^2$. Based on the grade of concrete, the allowable compressive stresses are 12.5N/mm² at transfer and 11.0N/mm² at service. The properties of the prestressing strands are: i) Type of prestressing tendon: 7-wire strand, ii) nominal diameter = 12.8 mm, iii) nominal area = 99.3 mm². Use preliminary design steps. (L3/CO4) 6

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